

Programming Project 1: Heuristic Query Optimization

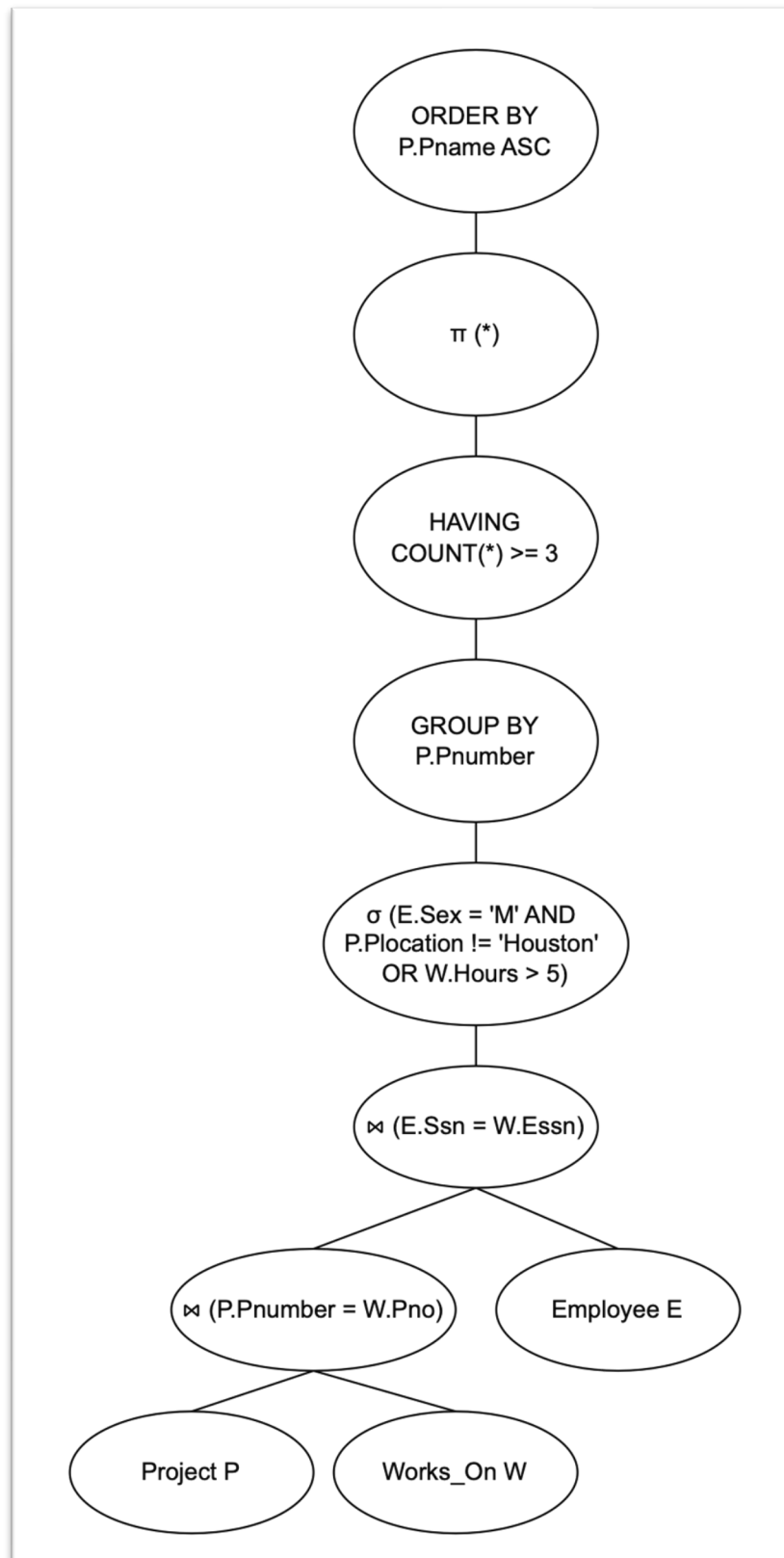
Example Inputs and Outputs

Example 1 (input2.txt)

Input SQL Query

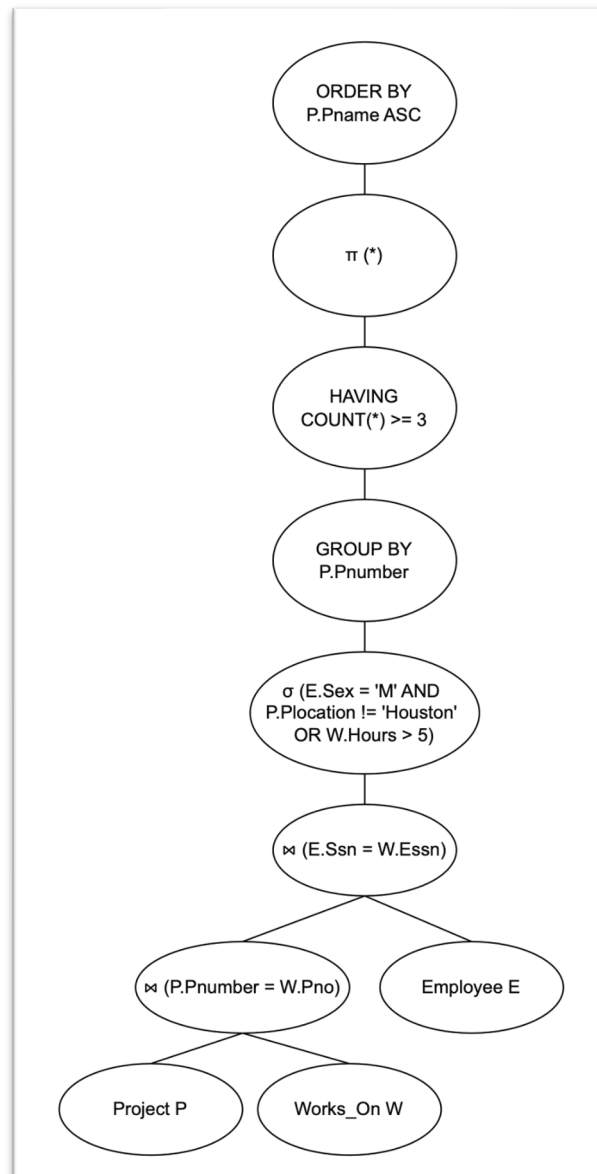
```
SELECT
  *
FROM
  Project P
  INNER JOIN Works_On W ON P.Pnumber = W.Pno
  INNER JOIN Employee E ON E.Ssn = W.Essn
WHERE
  E.Sex = 'M'
  AND P.Plocation != 'Houston'
  OR W.Hours > 5
GROUP BY
  P.Pnumber
HAVING
  COUNT(*) >= 3
ORDER BY
  P.Pname ASC;
```

Canonical Query Tree



Optimized Query Tree

Applied Rule #1 & #2: Identify conjunctive selections, break them up individually, and push them down as close as possible to the leaf nodes.



In this query tree example, the particular selection condition **cannot** be pushed down any further because it contains a **mixture** of AND and OR predicates. Therefore, we cannot break it into independent selections that apply cleanly to individual base relations. Pushing only parts of the predicate down would change the logical meaning of the query.

Applied Rule #3: Apply Selections with Smallest Selectivity First.

Not applicable.

Applied Rule #4: Replace Cartesian Product + Selection with Join

Not applicable.

Applied Rule #5: Push Projections Down

Not applicable.

